TechNova Customer-Support Chatbot

Hands-on assignment report | LangChain + LangGraph + NVIDIA NIM + LangSmith

1. Objective and implementation

The project implements a mock TechNova customer-support chatbot that answers order-status and policy questions. The workflow uses a ChatNVIDIA model for response generation, LangGraph to control the conversation flow, tools for order lookup and policy retrieval, and a MemorySaver checkpointer with thread IDs to preserve conversation context. LangSmith tracing was enabled under the project name **technova-bot** to inspect runs and retrieve token usage.

The mock order data includes TN1001 (Laptop Pro 14, shipped, ETA 2 days), TN1002 (Noise-cancel Earbuds, processing, ETA 5 days), and TN1003 (Phone X, delivered). Policy tools cover returns within 30 days in original packaging, a one-year manufacturer warranty, free shipping above Rs. 999, and standard delivery in 3–5 days.

2. Latency experiment

Metric	Measured result
Model	nvidia/nemotron-3.5-lightning-30b-a3b
Requests	20 total; 20 successful; 0 failed
Average latency	3.543 seconds
P50 latency	1.764 seconds
P95 latency	12.887 seconds

Interpretation: the median response completed in 1.764 seconds, while the P95 of 12.887 seconds shows that a slower tail exists even though all 20 requests succeeded. The mean is higher than the median, consistent with some slower calls. The recorded experiment measures end-to-end request latency; per-node latency and time-to-first-token (TTFT) results were not provided in the supplied summaries.

3. Token usage and estimated cost

Metric	Result
Model calls counted	94
Input tokens	46,780
Output tokens	5,908
Total tokens	52,688
Average tokens per call	560.51
Estimated small-model cost	$0.010538
Estimated large-model cost	$0.047419

The costs are hypothetical estimates using the assignment's rate card, not an actual NVIDIA invoice or billing record. The token summary includes 94 model runs collected from LangSmith; it should not be assumed to represent only the 20 latency-test requests.

Potential token optimizations include keeping the system prompt concise, returning only the necessary tool fields, limiting repeated conversation history where safe, and avoiding unnecessary tool calls. Any change should be re-tested for answer quality.

4. Error handling and functional evaluation

Five local fault/validation scenarios were marked Pass: empty order ID, unknown order ID, valid order ID, simulated tool failure, and timeout/retry behavior. The timeout test simulated an initial timeout followed by two retries, then a friendly failure. These were local tests and did not require a live API call; therefore, they demonstrate application-level handling, not measured live-service failure rates or proof that network retries work under actual provider outages.

The chatbot answered all 10 evaluation questions. A simple keyword/rule-based check marked 10/10 as Pass (100%). This is a preliminary heuristic and does not guarantee semantic correctness. The LLM-as-judge run produced complete scores for 6 of 10 questions (60% completion); for those six, average correctness, relevance, and completeness were each 5.0/5. Four questions had missing judge scores and must not be counted as scored or silently treated as perfect.

5. Recommendation, risks, and monitoring

Provisional model recommendation: keep *nvidia/nemotron-3.5-lightning-30b-a3b* as the tested baseline for this prototype because it completed all 20 measured latency requests. This is not a proven small-versus-large model winner: no comparable measured benchmark for a second model was included in the results supplied for this report. Before deployment, compare candidate models using the same questions, concurrency, prompt, and scoring rubric.

Top risks and proposed thresholds

- **Latency tail:** P95 was 12.887 seconds. Proposed alert if rolling P95 exceeds 10 seconds for 15 minutes; investigate slow model calls, tool latency, and prompt length.
- **Provider errors / rate limits:** 0 failures occurred in the 20-request latency run, but this sample is small. Proposed alert if request error rate exceeds 5% over a rolling 100 requests; use bounded retries with backoff and a friendly fallback.
- **Evaluation coverage:** only 6/10 LLM-judge scores completed. Proposed release gate: at least 90% of evaluation cases have valid judge scores, with critical order and policy cases reviewed for factual correctness.

Additional limitations: the supplied summaries do not include a measured second-model comparison, per-node timing, TTFT measurements, or a production error rate over 20 live requests. Those should be completed and recorded before making production-readiness claims.

6. Conclusion

The prototype demonstrates tool-based order and policy responses, thread-scoped conversation memory, LangSmith tracing, a 20-request latency test, token accounting, and local fault-handling checks. Initial functional results are promising, but the long-tail latency and incomplete LLM-as-judge coverage remain the main measurable gaps. A matched model comparison, TTFT/per-node instrumentation, and a larger live error test are recommended next steps.